ex200 - Exam Practice Questions

==========Question 1 ==========

Configure Network Settings

Configure the network interface `ens160` with the following settings:

- **IP Address**: `192.168.xxx.200/24`
- **Gateway**: `192.168.xxx.2`
- **DNS**: Use the default DNS settings.

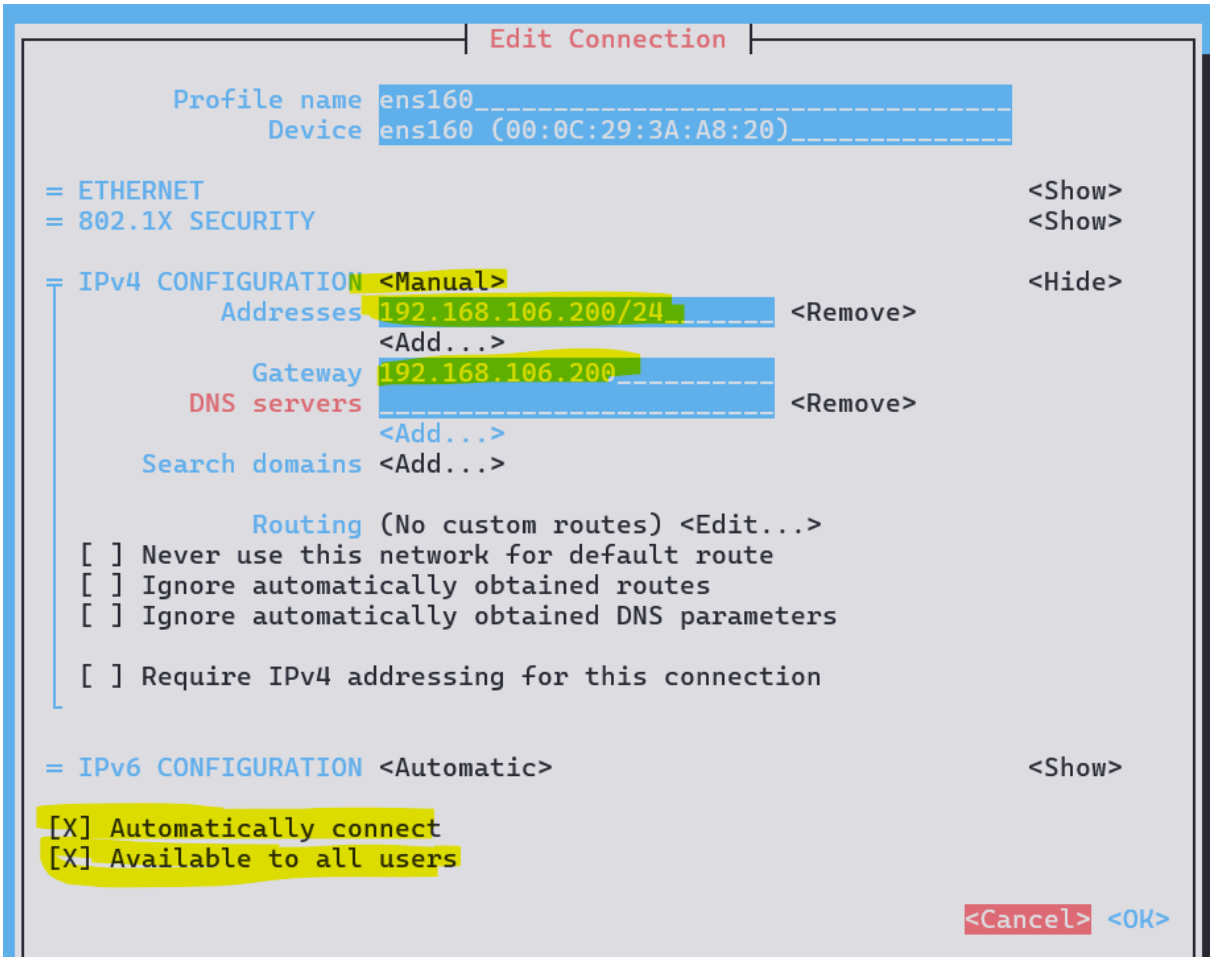
Ensure the changes are applied and the network interface is active.

==========Solution 1==========

#nmtui → edit connection → ens160 → make changes

#nmcli con down ens160

#nmcli con up ens160



==========Question 2 ==========

Configure DNF Repositories

Configure the system to use the following DNF repositories:

- **BaseOS:** http://classroom.example.com/content/RHEL9.0/x86_64/dvd/BaseOS
- **AppStream:** http://classroom.example.com/content/RHEL9.0/x86_64/dvd/AppStream

Ensure that GPG checking is disabled for these repositories.

==========Solution 2 ==========

vim /etc/yum.repos.d/package.repo

[BaseOS]
name=BaseOS

baseurl=
http://classroom.example.com/content/RHEL9.0/x86_64/dvd/BaseOS
gpgcheck=0
enabled=1

[AppStream]
name=AppStream
baseurl=
http://classroom.example.com/content/RHEL9.0/x86_64/dvd/AppStream
gpgcheck=0
enabled=1

#dnf clean all

#dnf makecache

#dnf repolist all

==========Question 3 ==========

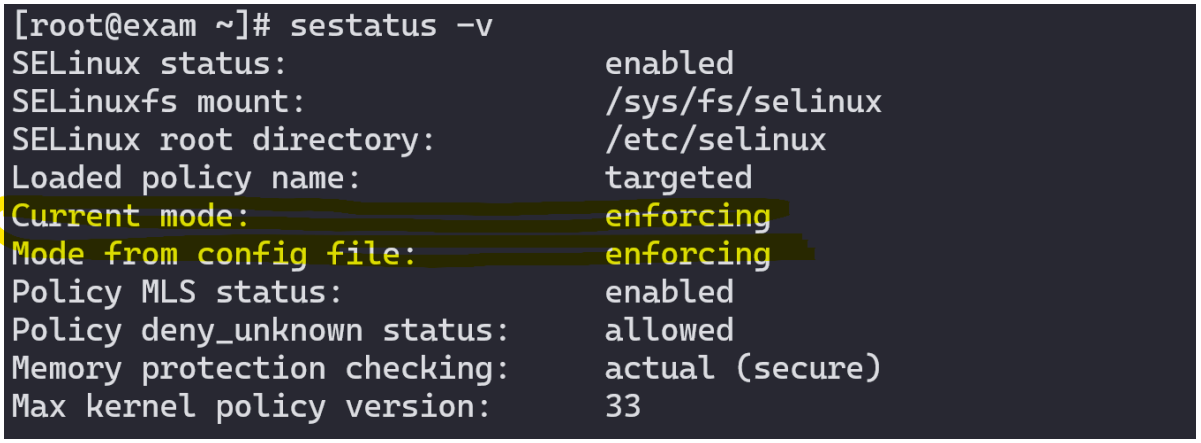
Troubleshoot Selinux

Install the `httpd` package and configure it to:

- Listen on port `82`.
- Serve a default webpage with the content `"Hello World"` located at `/var/www/html/index.html`.

Ensure that the service is enabled and started, and that the firewall allows traffic on port `82`.

==========Solution 3 ==========



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Enforcing

sestatus -v

dnf install httpd -y

systemctl enable --now firewalld

systemctl status httpd.service

systemctl enable --now httpd.service

vi /etc/httpd/conf/httpd.conf

Listen 8182

semanage port -a -t http_port_t -p tcp 8182

systemctl restart httpd.service

semange port --list

firewall-cmd --permanent --add-port=8182/tcp

firewall-cmd --reload

firewall-cmd --list-all

systemctl restart httpd.service

#cat “Hello World” > /var/www/html/index.html

Verify On BASE Machine

curl <server#1_ip>:8182

==========Question 4 ==========

Create Users and Groups

Create the following users and groups:

- A group named `sysmgrs`.
- A user `natasha` who belongs to `sysmgrs` as a secondary group.
- A user `harry` who belongs to `sysmgrs` as a secondary group.
- A user `sarah` who does **not** have access to an interactive shell and is **not** a member of the `sysmgrs` group.

Set the password for `natasha`, `harry`, and `sarah` to `secure123`.

==========Solution 4 ==========

#groupadd sysmgrs

#useradd sarah
#useradd natasha
#useradd harry

#usermod -G symgrs natasha
#usermod -G symgrs harry
#usermod -s /sbin/nologin sarah

#passwd sarah
#passwd natasha
#passwd harry

==========Question 5 ==========

Create a Collaborative Directory

Create a directory `/shared/sysmgrs` with the following characteristics:

- The group ownership of the directory should be `sysmgrs`.
- The directory should be readable, writable, and accessible only by members of the `sysmgrs` group.
- Files created in `/shared/sysmgrs` should automatically have their group ownership set to `sysmgrs`.

- Group ownership of /shared/sysmgrs is sysmgrs.
- The directory should be readable, writable, and accessible to member of sysmgrs, but not to any other user.
- Files created in /shared/sysmgrs automatically have group ownership set to the sysmgrs group.

==========Solution 5 ==========

#mkdir -p /shared/sysmgrs

#chown .sysmgrs /shared/sysmgrs

#chmod 2770 /shared/sysmgrs/

OR

#chmod g+rwx,o-rx /shared/sysmgrs

#chmod g+s /shared/sysmgrs

Verify

#ls -ld /shared/sysmgrs

==========Question 6 ==========

Configure a Cron Job

Configure a cron job for the user `natasha` that runs daily at `15:25` local time and executes the following command:

Copy

```
/bin/echo "Hello World"
```

Ensure the cron job is properly configured and verify its existence.

==========Solution 6 ==========

systemctl enable --now crond

crontab -e -u natasha

25 15 * * * /bin/echo “Hello World”

Verify

cat /var/spool/cron/natasha

==========Question 7 ==========

Configure Autofs for NFS Home Directories

Configure the system to automount users' home directories from an NFS server:

- The NFS server `nfsserver.example.com` (IP: `10.25.254.254`) exports the directory `/rhome`.
- When a user logs in, their home directory should be mounted from the NFS server to `/home/userX` (e.g., `nfsserver.example.com:/rhome/user10` → `/rhome/user10`).
- Ensure that home directories are writable by their respective users.

==========Solution 7 ==========

#dnf install autofs nfs-utils -y

#systemctl enable --now autofs.service

#vi /etc/auto.master.d/share.autofs

/rhome/ /etc/auto.netdir

#vi /etc/auto.netdir

* -rw,sync,fstype=nfs4 nfsserver.example.com:/rhome/&

systemctl restart autofs

Verfiy

su - user10

#df -h

==========Question 8 ==========

Configure NTP Client

Configure the system to synchronize time with the NTP server `time.example.com`. Ensure that the NTP service is enabled and started, and verify that the system is synchronizing time correctly.

==========Solution 8==========

systemctl enable --now chronyd

vi /etc/chrony.conf

server time.example.com iburst

timedatectl set-ntp true

systemctl restart chronyd.service

chronyc sources -v

Verify

timedatectl

==========Question 9 ==========

Resize a Logical Volume

Resize the logical volume `lv_test` from `192MB` to `250MB`. The logical volume is mounted on `/data` and uses the `ext4` filesystem. Ensure that the filesystem is resized accordingly and that the changes persist after a reboot.

==========Solution 9 ==========

#df -h

#lvs
#lvextend –L 250M /dev/vg_data/lv_test
#resize2fs /dev/vg_data/lv_test

Verify

#cat /etc/fstab
#df -h

==========Question 10 ==========

Create a Swap Partition

Create a swap partition of `512MB` on the disk `/dev/vdb`. Ensure that the swap partition is activated and configured to be available permanently after a reboot.

==========Solution 10 ==========

gdisk -l

gdisk /dev/vdb → n → partition no (Enter) → First Sector (Enter) →Last Sector +512M

→Hex code 8200 →p → w

partprobe /dev/vdb

#mkswap /dev/vdb3

swapon /dev/vdb3

#vim /etc/fstab

/dev/vdb3 none swap pri=10 0 0

mount -a

Verify

swapon -s

swapon -a

==========Question 11 ==========

Find and Copy Files

Find all files owned by the user `harry` and copy them to the directory `/root/findfile`.

==========Solution 11 ==========

#mkdir /root/findfile
#find / -user harry -print -exec cp -rp {} /root/findfile/ \;

==========Question 11-alternate ==========

Find all files owned by a user harry which are between 1MB and 5MB size and copy it into a /root/findfile directory.

==========Solution 11 ==========

#mkdir /root/findfile
#find / -user harry -size +1M -size -5M -print -exec cp -rp {} /root/findfile/ \;

==========Question 12 ==========

Write a Shell Script

Write a shell script that finds all files owned by the user `harry` and copies them to the directory `/root/findfile`. Ensure the script is executable and test it to verify its functionality.

==========Solution 12 ==========

#vi script.sh
#!/bin/bash
mkdir /root/findfile
find / -user harry -exec cp -rp {} /root/findfile/ \;

#chmod +x script.sh
#./script.sh

#ls -l /root/findfile

==========Question 13 ==========

Search and Save Lines

Search the file `/usr/share/dict/words` for all lines containing the string `"strato"` and save them to a file named `files.txt` in the `/root` directory.

==========Solution 13 ==========

#grep strato /usr/share/dict/words
#grep strato /usr/share/dict/words > /root/files.txt

==========Question 14 ==========

Create a User with a Specific UID

Create a user named `manalo` with the user ID `5233`. Set the password for `manalo` to `redhat`.

==========Solution 14 ==========

#useradd -u 5233 manalo
#passwd manalo

==========Question 15 ==========\

Create a Compressed Archive

Create a compressed archive named `backup.tar.bz2` of the `/usr/sbin` directory. Ensure the archive is created in the root (`/`) directory and is compressed using the `bzip2` method.

==========Solution 15 ==========

#cd /
#tar -cvjf backup.tar.bz2 /usr/sbin

==========Question 16 ==========

Reset the Root Password

Reset the root password. Ensure that the system remains bootable and that SELinux contexts are properly relabeled after the password reset.

==========Solution 16 ==========

reboot vm2

on grub menu →go to rescue kernel and press "E"

make sure **rd.break console=tty1** is is the first arg in the line for linux kennel

CTRL+X

mount -o remount,rw /sysroot

chroot /sysroot

passwd root

touch /.autorelabel

exit

exit

==========Question 17 ==========

Create a Logical Volume

Create a logical volume named `storage` with a size of `20 PE` (Physical Extents) by creating a volume group named `redhat` where `1 PE = 32MB`. Format the logical volume with the `vfat` filesystem and mount it on `/mnt/data2`. Ensure the mount is persistent across reboots.

==========Solution 17 ==========

gdisk /dev/vdb → n → First Sector → Last Sector +850M → Hex Code → 8e00 →w

#partprobe

#pvcreate /dev/ vdb8 (to make the partitions as physical volume.
#vgcreate -s 32M redhat /dev/vdb8
#lvcreate -n storage -l 20 redhat

#mkfs.vfat /dev/redhat/storage
#mkdir -p /mnt/data2

#mount /dev/redhat/storage /mnt/data2
#vim /etc/fstab
/dev/redhat/storage /mnt/data2 vfat defaults 0 0

Verify
#mount -a
#df -h

==========Question 18 ==========

Configure a VDO Volume

Create a VDO volume named `vdo1` with a size of `5GB` on a new disk. Format the volume with the `xfs` filesystem and mount it on `/database`. Ensure the mount is

persistent across reboots.

==========Solution 18 ==========

#gdisk /dev/vdb → n → First Sector → Last Sector +5G → Hex Code → 8e00 →w

#partprobe

#pvcreate /dev/vdb4

#vgcreate vg_vdo /dev/vdb4

#lvcreate --type vdo --name vd01 -L 5G vg_vdo
#mkfs.xfs -K /dev/mapper/vg01-vd01
#vi /etc/fstab

#mkdir /database
/dev/mapper/vg01-vd01 /database   xfs defaults 0 0

#mount -a

Verify

df -h

==========Question 19 ==========

Set Tuned Profile

Set the system's tuned profile to `virtual-guest` or the recommended profile. Ensure the tuned service is enabled and running.

==========Solution 19 ==========

#systemctl enable --now tuned

#tuned-adm active

#tuned-adm profile

#tuned-adm recommend

or

#tuned-adm profile virtual-guest

==========Question 20 ==========

Build and Run a Container

As the user `neith`, perform the following tasks:

1. Log in to the container registry at `registry.network14.example.com`.
2. Download the `Containerfile` from the following URL:

   Copy

   ```
   https://github.com/abhatt014/rhcsa/blob/main/Containerfile
   ```

3. Build a container named `watch` using the `Containerfile`.
4. Map the local directory `/opt/in` to the container's `/opt/incoming` and `/opt/out` to the container's `/opt/outgoing`.
5. Run the container as a service named `ascii2pdf` and ensure it is enabled to start automatically.

==========Solution 20 ==========

#ssh root@server

#usermod -G wheel neith

#passwd neith

logout as root

#ssh neith@server

#wget https://github.com/abhatt014/rhcsa/blob/main/Containerfile

#ls -l

#sudo dnf install container-tools

#podman login resigtry.network14.example.com

#podman build -f Containerfile -t watch

#podman images

#sudo loginctl enable-linger neith

#cd /opt ; mkdir in out

#chmod 777 in ; chmod 777 out

#chown neith:neith in ; chown neith:neith out

#podman run -d --name ascii2pdf -v /opt/in:/opt/incoming:Z -v /opt/out:/opt/outgoing:Z localhost/watch:latest

#podman ps

#mkdir -p ~/.config/systemd/user ; cd ~/.config/systemd/user

#podman generate systemd --name ascii2pdf --files

#systemctl --user daemon-reload

#systemctl --user status container-ascii2pdf. service

#systemctl --user enable container-ascii2pdf. service --**now**